Quantum Entanglement

Quick Recap

• A qubit may exist in a superposition of |0⟩ and |1⟩.

• The Hadamard (H) gate creates superposition.

• The CNOT gate lets one qubit control an operation on another.

• Two qubits are described together as one quantum system.

1. What Is Quantum Entanglement?

Quantum entanglement occurs when two or more qubits share a quantum state that cannot be fully described by treating each qubit separately.

• The qubits are described by one combined state.

• Their measurement outcomes can be strongly correlated.

• Measuring one qubit does not send an instant message to the other.

Everyday analogy: Two synchronized clocks show a relationship, but entanglement is stronger than ordinary classical synchronization. The clocks are only a memory aid.

2. Classical Correlation vs Quantum Entanglement

Classical correlation	Quantum entanglement
Objects have their own states.	The qubits share a combined quantum state.
Correlation can arise from shared history.	The shared state requires quantum mechanics.
A classical explanation is possible.	Not fully explained by classical physics.

3. How to Create an Entangled Pair

Begin with two qubits in |00⟩. Apply H to the first qubit, then CNOT with the first qubit as control and the second as target.

```
Initial |00>  ->  H on q0  ->  CNOT(q0,q1)  ->  Bell state
```

Real-life analogy: Think of a two-step recipe: prepare the first ingredient, then connect the pair. Both steps are needed for this particular Bell-state circuit.

4. The First Bell State

The resulting state is the Bell state $|\Phi+\rangle = (|00\rangle + |11\rangle) / \sqrt{2}$.

• Only $|00\rangle$ and $|11\rangle$ appear in ideal computational-basis measurements.

• Each of these outcomes has a probability of 50%.

• The next outcome is random, but the two measured bits agree.

Measurement	Ideal probability	Illustrative counts (1,000 shots)
00	50%	501
01	0%	0
10	0%	0
11	50%	499

Figure 1. Ideal Bell-state outcomes: only 00 and 11 occur.

5. Circuit Representation

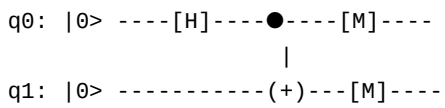


Figure 2. H creates superposition; the CNOT (● and ⊕) creates the entangled Bell pair.

6. Superposition Is Not the Same as Entanglement

• Superposition can describe one qubit.

• Entanglement describes a combined state that cannot be separated into independent qubit states.

• A CNOT gate does not always create entanglement; the input state matters.

7. Qiskit Example

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(2, 2)
qc.h(0)
qc.cx(0, 1)
qc.measure([0, 1], [0, 1])
print(qc.draw())
```

This code constructs and displays the circuit. To obtain measurement counts, it must also be executed using a simulator or quantum backend.

Expected ideal results: approximately 50% 00 and 50% 11. Real hardware may show other results because of noise and decoherence.

8. Why Entanglement Matters

• Quantum teleportation and quantum communication.

• Quantum cryptography and networking.

• Superdense coding and quantum error correction.

• Some quantum algorithms and quantum machine learning techniques.

Example: Researchers study entangled photons for quantum communication. The workbook introduces quantum key distribution (QKD) as a related secure-communication application.

9. Common Beginner Mistakes

• “Entangled qubits always have opposite values.” No. The Bell state studied here produces matching values.

• “Entanglement sends messages faster than light.” No. Correlation is not instant controllable communication.

• “Superposition and entanglement mean the same thing.” No. One describes possible states; the other describes a non-separable shared state.

• “CNOT alone always creates entanglement.” No. The input must be suitable.

10. Key Formula and Visual Recap

```
        |00> + |11>
|Φ+> = ------------
           sqrt(2)

|00> -> H on first qubit -> CNOT -> |Φ+>
                                 -> Measure: 00 or 11
```

Figure 3. Bell-state preparation and measurement in one flow.

Summary: What to Remember

• Entanglement belongs to the combined quantum system.

• H followed by CNOT can create the Bell state |Φ+⟩.

• In ideal measurements, |Φ+⟩ produces 00 or 11 with equal probability.

• Entanglement is different from ordinary classical correlation.

• It is useful in quantum communication, cryptography, networking and algorithms.

Resources Mentioned in the Workbook

Real Python – Quantum Computing Basics; Quantum Entanglement Explained (video); and a video walkthrough of entangled circuits. See the original Day 10 workbook for the links.